MoonViT and Native Multimodality

Deep Dive into Kimi K3 — Chapter 9

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Foundation Books Project

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Goal: realize budgeted grids, pack them without losing position or isolation, encode, reduce, project, splice, and keep each source's authority in scope.

Roadmap

Grids, packing, and position

Encode, reduce, project, splice

Ledgers, lineage, and diagnosis

Grids, packing, and position

From pixels to isolated patch rows



- Native resolution is a budgeted family of grids; padding is counted after the scale choice.

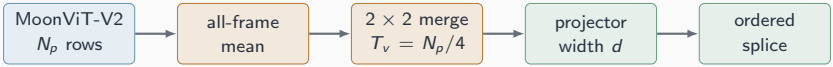
From pixels to isolated patch rows



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- Packing keeps offsets, local coordinates, and a block-diagonal mask; position signals and boundaries have different jobs.

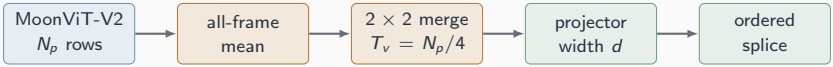
Encode, reduce, project, splice

From encoder rows to one shared stream



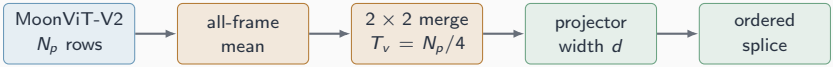
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- Splice by marker identity, not by width; count $T = T_x + T_c + \sum T_{v,i}$.

Ledgers, lineage, and diagnosis

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Diagnose in construction order. Next: Chapter 10's Integrated K3 Transformer.